# Chapter 5 - MEMORY

**Memory Hierarchy** = A structure that uses multiple levels of memory.

(Distance from processor ↑ → Size of memories & access time ↑, Cost per bit ↓)

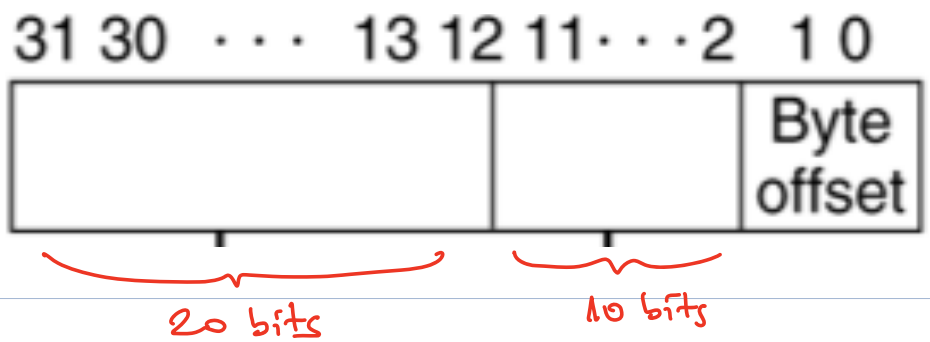
Block (or line) = minimum unit of information that can be present or not present in cache.

Hit Rate = Fraction of memory accesses found in a level of the memory hierarchy.

**Direct-Mapped Cache** = A cache structure in which each memory location is mapped to exactly one location in the cache.

Tag = A field in a table used for a memory hierarchy that contains the address info. required to identify whether the associated block in the hierarchy corresponds to a requested word.

| Decimal address of reference | Binary address of reference | Hit or miss in cache | Assigned cache block (where found or placed) |
|---|---|---|---|
| 22 | $10110_{two}$ | miss (5.9b) | $(10110_{two} \bmod 8) = 110_{two}$ |
| 26 | $11010_{two}$ | miss (5.9c) | $(11010_{two} \bmod 8) = 010_{two}$ |
| 22 | $10110_{two}$ | hit | $(10110_{two} \bmod 8) = 110_{two}$ |
| 26 | $11010_{two}$ | hit | $(11010_{two} \bmod 8) = 010_{two}$ |
| 16 | $10000_{two}$ | miss (5.9d) | $(10000_{two} \bmod 8) = 000_{two}$ |
| 3 | $00011_{two}$ | miss (5.9e) | $(00011_{two} \bmod 8) = 011_{two}$ |
| 16 | $10000_{two}$ | hit | $(10000_{two} \bmod 8) = 000_{two}$ |
| 18 | $10010_{two}$ | miss (5.9f) | $(10010_{two} \bmod 8) = 010_{two}$ |
| 16 | $10000_{two}$ | hit | $(10000_{two} \bmod 8) = 000_{two}$ |




The cache holds 1024 words = 4 KiB

Block size = 1 word & cache has $1024 = 2^{10}$ words

(10-Bit for index 32 − 10 − 2 = 20 bit for TAG)

Block = (Block Addr) mod (# of blocks in cache)

Block Addr = Byte Addr / Bytes per block

(Block Addr) is the block containing all addresses between

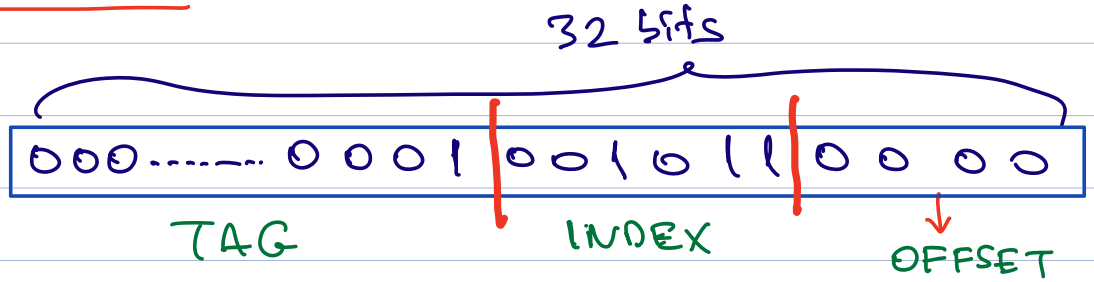
$$\left[\frac{\text{byte addr}}{\text{bytes per block}}\right] \times \text{Bytes per block}$$

$$\left[\frac{\text{byte addr}}{\text{bytes per block}}\right] \times \text{Bytes per block} + (\text{Bytes per block} - 1)$$

**Question:** Address → 1200 mapped to which block?

Assume 64 blocks & 1 block = 16 bytes given

**Answer**




Index = 11
(Block number)

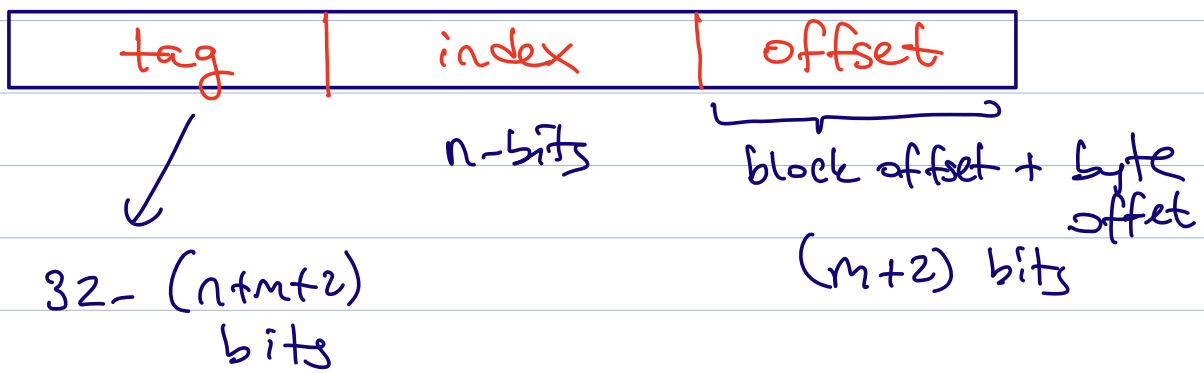
That block maps all address between 1200 and 1215

## Total # of Bits for a Cache

Total # of bits needed for a cache is a function of cache size & address size

* 32-bit address
* Direct-mapped cache
* Cache size is $2^n$ blocks (n-bit for index)
* Block size is $2^m$ words ($2^{m+2}$ bytes)
  (m-bits word within block, 2 bits byte part of the addr)

Size of tag field: $32-(n+m+2)$



Total # of bits in a direct-mapped cache

$= 2^n \times$ (block size + tag size + valid field size)

$= 2^n \times (2^m \times 32 + (32 - n - m - 2) + 1)$

$= 2^n \times (2^m \times 32 + 31 - n - m)$

